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Research on application of flowforms in combination with planted constructed wetland for improving water quality of urban polluted lakes

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Abstract. With continuously increasing urbanization in Vietnam, urban lakes pollution by domestic wastewater has become a serious issue, which results in degradation of lake water quality and consequent deterioration of landscape as well as environment for urban residents. Finding low-cost, simple and sustainable solutions would help to overcome that issue in Vietnam's context. This study aims at proposing a solution which includes low-cost, nature- and landscape-friendly processes such as wetlands and flowforms for improving polluted urban lakes. The study was carried-out on a pilot system consisting of a cascade of flowforms and a planted constructed wetland with flow recirculation. Water samples used in this study have been collected from an urban lake in Hanoi, which receives wastewater from a densely populated residential area. The research results have demonstrated that this solution could achieve high treatment efficiencies: Color - 92%, Chemical oxygen demand (COD) - 92%, Ammonia (NH_4^+-N) - 98% and Total Phosphorus (TP) - 87%. The effluent quality could meet requirement by Vietnam's National technical regulation on surface water quality QCVN 08-MT:2015/BTNMT, level B1 in terms of the above mentioned parameters.

Keywords: Urban, lake, pollution, water, landscape, flowform, wetland, low-cost.

1 Introduction

Recently in Vietnam, the utilization of low-cost or simulating nature processes such as wetlands and flowforms for improving urban water environment has drawn an increasing interest due to their simplicity, sustainability and friendly landscape.

Planted constructed wetland (CW) is a system of vegetation, soil, water and microbial populations that uses natural processes to carry out water treatment. The processing takes place in CWs largely due to decomposition by microorganisms along with plant uptake to remove organic matter, nitrogen, etc... Amongst the factors influencing CWs performance, dissolved oxygen (DO) plays a crucial role, since it is vital for microbial activities [1]. It has been consistently documented in many reports that the higher the oxygen content the faster organic matter can be degraded in CWs [2]. Besides, due to

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competition for oxygen between organic matter decomposition and nitrification processes, lack of DO would inhibit nitrification or even affect the total nitrogen removal efficiency to a large extent [3]. Amongst the solutions for enhancing DO content in wastewater engineering, flowforms have emerged as an effective, low-cost and landscape-friendly option which has been widely applied in Europe, US and Australia [4].

Flowforms are vessels that seek to emulate the swirls of vortices of the mountain stream enabling water to re-oxygenate (Fig. 1). When arranged in a cascade, flowforms would provide strong vortices resembling a figure of "8" (Fig. 1) and help the aeration to occur effectively in a small space, while creating a harmonious movement of water that simulates nature rhythms.

This paper presents the results of the research on application of flowforms (Fig. 2) in combination with a horizontal subsurface flow CW (HSFCW) for treatment of urban lake water polluted by domestic wastewater.

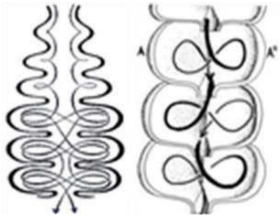

Fig. 1. Flow patterns in flowforms [4]

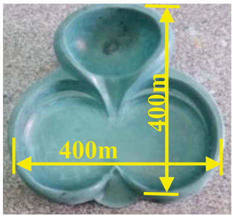

Fig. 2. Flowform shape

2 Materials and methods

2.1 Materials

The pilot system for experiments is as shown in Fig. 3 and includes the following components:

- *Flowforms cascade:* consists of 7 vessels made of ultra-high performance concrete with dimensions 400 x 400mm in plan, 140mm in height and shape as shown in Fig. 1.

- *HSFCW:* dimensions B×L×H = 0,55×1,2×0,4m. Crushed stone was used as media and arranged into 3 layers. Bottom layer: 10cm in thickness and 3,5÷5cm in size; middle layer: 15cm in thickness and 1,5÷3cm in size; top layer 15cm in thickness and 0,5÷0,7cm in size. The water depth in the CW was maintained at 35cm. 20L of thickened activated sludge from a local domestic wastewater treatment plant was used for seeding at the acclimatization period. Umbrella papyrus (*Cyperus alternifolius),* a ubiquitously found specie in Vietnam, has been selected to plant at a density 100 trees/m^2. The plants with the height ranged from 30cm to 60cm were taken from the same lake where water samples were collected in order to facilitate their acclimatization. The acclimatization period lasted 15 days with the inlet flow of 150L/h. From the 7^{th} day, the growth in length of the plants at 1cm/d÷1,5cm/d were observed.

- *Storage tank:* dimensions B × L × H = 0,5 × 1,2 × 0,5m;

- *Flow stabilizator:* dimensions D x H = 190 x 500mm, made of PVC and used for maintaining a constant flow to the experimental setup;

- *Equipment*: a pump used for feeding the experimental system; the valves V1, V2, V3, V4 used for regulating the influent rate, pump flow rate and drainage respectively.

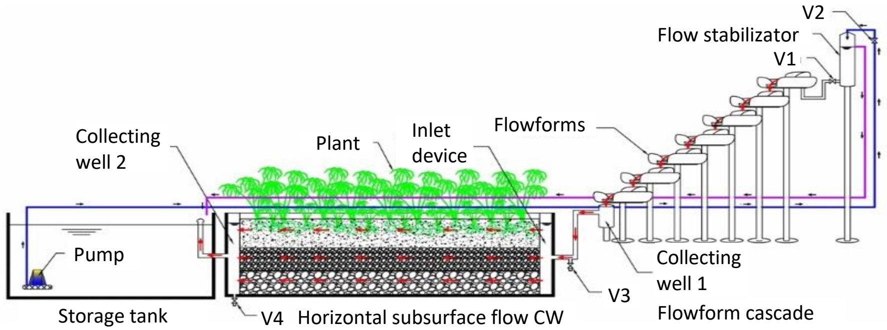

Fig. 3. Pilot system components

2.2 Methods

Experimental setup

Lake water samples were fed into the experimental setup with recirculation flow as follows: Storage tank → Flowforms → HSFCW → back to Storage tank.

The aeration performance of the flowform as shown in Fig.1 has been previously studied by the authors of this paper [5]. The highest aeration rates were obtained at flow rates ranging between 150L/h and 200L/h. Therefore, the flow rates of 150L/h (in 1^{st} stage) and 200L/h (in 2^{nd} stage) were chosen for feeding the pilot system. The experiment lasted from March 20 to May 27, 2018 with three batches of 10 days in each stage.

Sampling and data analysis

Lake water samples were collected from Kim Lien lake (Hanoi) – a typical urban lake, polluted by domestic wastewater. Samples quality parameters are shown in Table 1.

Samples were collected from Storage tank once a day and analyzed for pH, Color, COD, NH_4^+-N, NO_3^--N, TP in accordance with standard methods [6]. At each stage, the results of 3 batches were averaged and presented on the Table 1 below.

Table 1. Quality parameters of lake water samples

Parameter	Stage 1			Stage 2			QCVN 08-MT:2015/ BTNMT, level B1
	Spl 1	Spl 2	Spl 3	Spl 4	Spl 5	Spl 6	
Temperature (°C)	29,7	28,9	31,2	26,5	28,3	26,3	-
pH	7,0	6,8	6,9	6,9	6,7	7,1	5,5-9
COD (mg/L)	166,6	174,9	181,3	279,2	248,5	165,7	30
DO (mg/L)	0,2	0,3	0,2	ND	ND	ND	≥ 4
NO_3^--N (mg/L)	0,15	0,09	0,06	0,06	0,12	0,05	10
NH_4^+-N (mg/L)	32,6	30,2	29,4	23,6	24,2	26,8	0,9
TP (mg/L)	1,57	1,62	1,71	2,56	2,48	2,56	PO_4^{3-}-P ≤ 0,3
Color (Pt-Co)	308,3	304,4	276,3	450,5	350,0	299,6	-

Remark: Spl – Sample; ND – Non-detectable.

DO concentration before and after CW was measured every hour (from 7:00 to 17:00), daily average values were used for assessment of DO change through CW.

Removal efficiencies of the experimental setup were calculated using formula:

$$E = \frac{C_0 - C_1 \frac{V_0 \; - V_{lost}}{V_0}}{C_0} \times 100\% \qquad (1)$$

Where C_0: initial concentration (mg/L); C_1: concentration at the end of each run (mg/L); V_0: initial volume of lake water sample (L); V_{lost}: volume of water being lost by sampling, evaporation and transpiration (L).

3 Results and Discussion

3.1 Dissolved oxygen change in CW

Fig. 4 below denotes that after the first day of operation, average DO concentration at the CW inlet reached 3,9mg/L to 4,8mg/L and remained stable throughout each stage. This has proved the high aeration efficiency of the flowforms. The DO value at the outlet from CW fluctuated from 2,9mg/L to 3,7mg/L, which indicates that there was sufficient DO for both organic matter decomposition and nitrification to occur in CW.

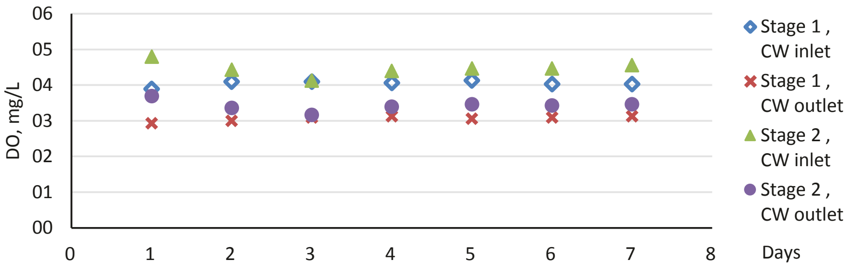

Fig. 4. DO change in CW

3.2 Performance of the experimental setup

Data on Fig. 5 and Fig. 6 indicate that, although the influent rates at each stage of experiment were significantly different (150L/h in the 1st compared to 200L/h in the 2nd), performances in both stages followed a similar trend.

COD removal: during both stages, a sharp decrease of COD concentration in the first 4 days and slower decline from the 5th day was observed. On the 7th day, COD concentration reached 28,1mg/L in 1st stage and 26,1mg/L in 2nd stage. Removal efficiencies were as high as: 90,7% in 1st stage and 92,6% in 2nd stage. Such trend shows that biodecomposition was the main mechanism of COD removal, although filtration through media layers and accumulation in root zone should have also take place.

NH_4^+-N removal: as can be seen on Fig. 5 and Fig. 6, there was a consistency in changes of NH_4^+-N and NO_3^--N concentrations. During the first 4 days of operation,

NH_4^+-N content decreased by more than 10 times, down to 1,9mg/L and 1,7mg/L in 1st and 2nd stages respectively, while NO_3^--N content rose sharply from 0,1mg/L to 5,5mg/L in 1st stage and from 0,08mg/L to 4,3mg/L in 2nd stage. After the 7th day, the average ammonia removal efficiency could reach 98% in both stages with NH_4^+-N content as low as 0.8mg/L in 1st stage and 0,4mg/L in 2nd stage, while NO_3^--N content remained less than 5,6mg/L in 1st stage and 4,5mg/L in 2nd stage.

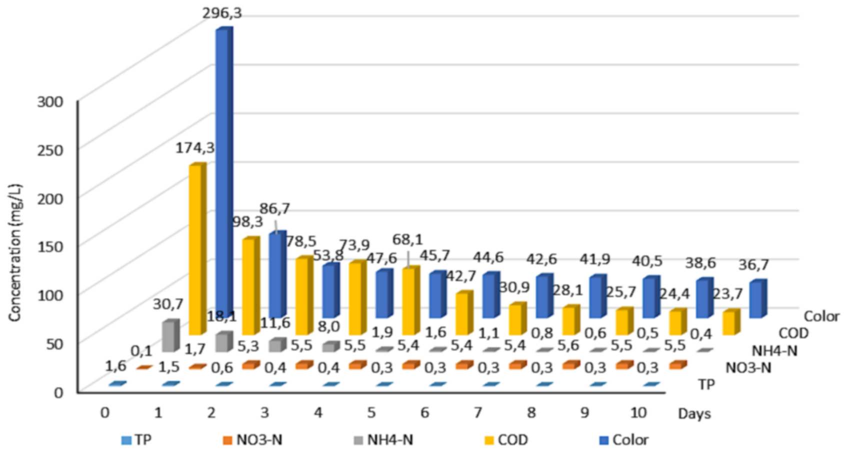

Fig. 5. Performance of the experimental setup during the 1st stage

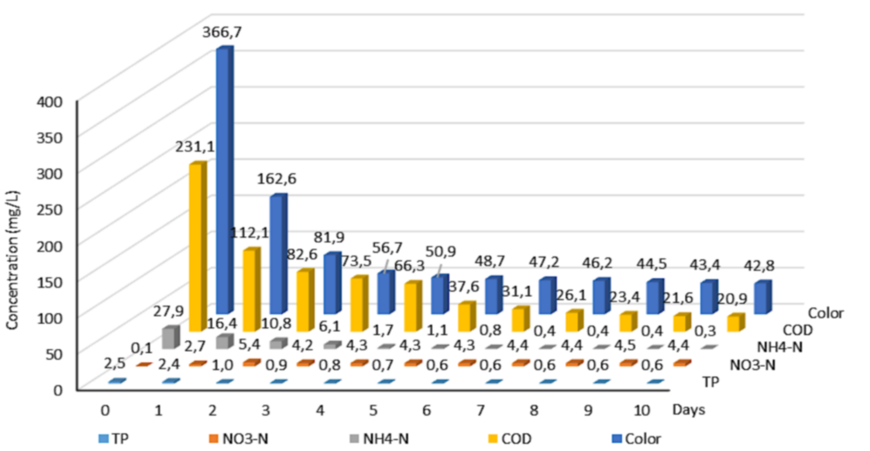

Fig. 6. Performance of the experimental setup during the 2nd stage

While the DO data characterized the aerobic condition in CW during both stages, the data of NH_4^+-N indicate that such low concentrations of ammonia were the result of the nitrification process. Since denitrification process was not likely to happen, the low content of NO_3^--N concentrations would link mostly to plant uptake.

Compared to the results of Li et al. [7], both COD and NH_4^+-N removal efficiencies after 7th day were at the same level – about 90% and 98% respectively. This points out that, although being a passive and simple device, the flowforms can perform as good as air compressor - aeration tubes system used in study by Li et al. [7].

TP removal: TP decreased rapidly in the first 3 days in both stages, then gradually decreased to 0,3mg/L in 1st stage and 0,6mg/L in 2nd stage. Such trend of TP content would be described that it was removed mainly by accumulation in filter media over accumulation by aerobic microorganisms and plant uptake. The TP removal efficiency reached 89% in 1st stage 1 and 85% in 2nd stage.

Color removal: data show a very good removal efficiency at about 92% and a stable level at about 40Pt-Co from the 7th day of the operation in both stages.

The experiment results have demonstrated very good performance of the Flowforms - HSFCW pilot system in treatment of polluted lake water samples: after 7 days of operation, all the parameters COD, NH_4^+-N, NO_3^--N and TP (in terms of PO_4^{3-}) would meet the requirement by QCVN 08-MT:2015/ BTNMT, level B1 (see Table 1 above).

4 Conclusions

The initial results proved that although being very simple, flowforms are a very effective option for aeration. Coupled with flowforms, CW systems could deliver high performance in removing organic matters, nutrients and color, thus could be applied for improvement of urban lakes, polluted by domestic wastewater. These systems could be placed next to or integrated into the lakes embankment and would serve not only enhancing water quality, but also shaping lake landscape, especially in Vietnam's context.

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References

1. Wu H., Zhang J., et all: A review on the sustainability of constructed wetlands for wastewater treatment: Design and operation. Bioresource Technology, 175, 594-601 (2015).
2. Pan J., Fei H., Song S., et all: Effects of intermittent aeration on pollutants removal in subsurface wastewater infiltration system. Bioresource Technology, 191, 327-331 (2015).
3. Hu Y., Zhao Y., et al.: High rate nitrogen removal in an alum sludge-based intermittent aeration constructed wetland. Environmental Science & Technology, 46(8), 4583-4590 (2012).
4. Schwuchow J., Wilkes J., Trousdell I.: Energizing water – Flowform technology and the power of nature, Sophia Books (2010).
5. Ung T. T. H., Leu T. B. et al.: Research on application of flowforms to improve water quality and create environment-friendly landscapes at polluted lakes. Grant No134-2017/KHXD-TD. National University of Civil Engineering (2018).
6. APHA, AWWA, WEF: Standard methods for the examination of water and wastewater. 19th Edition, American Public Health Association, Washington, D.C., USA (1995).
7. Li F., Lu L., Zheng X., et all: Enhanced nitrogen removal in constructed wetlands: Effects of dissolved oxygen and step-feeding. Bioresource Technology, 169, 395-402 (2014).